



Data Article

DeepUS-ReconSeg: A multi-angle paired B-mode Ultrasound dataset for medical imaging reconstruction and segmentation

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ARTICLE INFO

Article history:

Received 23 June 2025

Revised 3 September 2025

Accepted 12 September 2025

Available online 19 September 2025

Dataset link: [Ultrasound high-resolution \(HR\) and low-resolution \(LR\) Hand arm Tissue image data \(B-mode\) \(Original data\)](#)

Keywords:

Ultrasound imaging

B-mode data

Verasonics Vantage

Linear array transducer

Medical dataset

Forearm ultrasound

ABSTRACT

This article presents a curated dataset of 4,200 B-mode ultrasound images of human forearms, intended for use in deep learning-based image reconstruction and segmentation tasks. Data have been collected using the Verasonics Vantage 64LE system equipped with an L11-5v linear array transducer, known for its high spatial resolution. Each scanning session captures 100 frames per orientation across both arms of 14 healthy subjects, covering multiple anatomical views. The dataset provides grayscale B-mode images stored in MATLAB .mat format and is publicly available through Mendeley Data. This dataset is valuable for researchers in medical image analysis, especially those developing deep learning models for enhancing ultrasound imaging quality and anatomical structure segmentation.

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Specifications Table

Subject	Medical AI, Deep Learning, Image Reconstruction
Specific subject area	Biomedical Ultrasound Imaging
Type of data	Filtered measurement data is stored as .zip (.mat) files.
Data collection	Data has been acquired using a Verasonics Vantage 64LE ultrasound system with an L11-5v linear array transducer. Each subject underwent scanning on both forearms in two orientations (longitudinal and transverse), capturing 100 frames per scan orientation.
Data source location	Pusan National University, Busan, South Korea
Data accessibility	Repository name: Mendeley data Data identification number: doi: 10.17632/tgbjc4rcg2.1 Direct URL to data: https://data.mendeley.com/datasets/tgbjc4rcg2/1
Related research article	None

1. Value of the Data

- This dataset provides a valuable resource for developing and evaluating deep learning models focused on ultrasound image enhancement, super-resolution, and structural reconstruction. It includes carefully paired high-resolution (HR) and low-resolution (LR) B-mode images, enabling precise training and benchmarking.
- It is particularly useful for researchers working on ultrasound-based artificial intelligence, image reconstruction, speckle noise reduction, and tissue segmentation. The dataset supports experimentation under varying imaging conditions and anatomical diversity.
- The inclusion of multiple probe steering angles (ranging from 1° to 75°) simulates realistic handheld scanning scenarios, enhancing the generalizability of AI models for clinical applications.
- Although the data were acquired from a limited group of healthy adult volunteers, the wide angular range and consistent scanning protocol introduce meaningful diversity, making the dataset suitable for training, fine-tuning, and validating machine learning models, particularly in early-stage or exploratory studies.
- The dataset follows a well-structured organization and standardized acquisition process. This ensures ease of use for both novice researchers and experienced developers working on tasks such as tissue boundary detection, resolution mapping, and image quality improvement.
- The dataset is well-suited for evaluating and benchmarking deep learning architectures such as TransCycleGAN [1], U-Net [2], and other encoder-decoder models intended for ultrasound image translation and semantic segmentation.

2. Background

This dataset has been generated to support the design and evaluation of deep learning models for ultrasound image enhancement and segmentation, particularly in cases where data are acquired under suboptimal imaging conditions. Using the programmable Verasonics Vantage 64LE system, B-mode images are captured at various aperture angles to simulate both high- and low-resolution scenarios common in clinical practice. The target anatomy—human forearm soft tissue—is selected due to its structural complexity and relevance to musculoskeletal and vascular imaging. The primary motivation is to generate a consistent, IRB-approved, and well-organized dataset suitable for training and benchmarking AI-based reconstruction and diagnostic tools that require paired or variable-resolution ultrasound data.

3. Data Description

The dataset comprises 4,200 grayscale B-mode ultrasound images, organized into 42 MATLAB .mat files [3], with each file containing 100 frames of size 256 × 256 pixels. These files represent two main categories:

- **High-Resolution (HR) images:** have been acquired using wide probe aperture angles (e.g., 75°) or high-frequency transducer settings (>15 MHz). These frames exhibit clearer boundaries and well-defined anatomical features such as muscle, fat, and skin interfaces.
- **Low-Resolution (LR) images:** have been captured at narrow probe angles (1°–15°) or generated by synthetically downsampling HR frames to mimic common degradation in clinical ultrasound imaging.

All images are provided in grayscale format and stored in .mat containers compatible with standard medical image processing tools and deep learning frameworks. The complete dataset occupies approximately 1.97 GB and is fully anonymized, with files named according to acquisition parameters and subject IDs for clarity. The specifications of the ultrasound system and transducer used during data acquisition are summarized in [Table 1](#).

Table 1
Ultrasound transducer specifications.

Machine	Verasonics vantage system 64 LE
Transducer	L11-5v (1D array)
Transducer Spacing	0.3mm
Transmit Wave	Plane Wave (zero-angle)
Interpolation	4pt
Center Frequency	7.6MHz
Acquisition FPS	60Hz
Pulser Voltage	11.3V
Bandwidth	[4.6800, 10.5200] MHz

4. Experimental Design, Materials and Methods

4.1. Data visualization

To ensure the integrity and interpretability of the acquired ultrasound data, we performed both raw and processed visual assessments.

- **Raw B-mode Visualization:** The original image frames extracted from each .mat file are visualized directly in grayscale using MATLAB and Python tools as shown in [Fig. 1](#). These raw images, stored as 3D arrays, exhibited a wide dynamic range with visually dark regions due to the presence of low-amplitude echoes. This baseline visualization was used to confirm frame structure, anatomical consistency, and acquisition quality across subjects and scan orientations.
- **Processed Visualization with Log Compression and Hilbert Transform:** To enhance interpretability and bring out soft-tissue details, we have applied log compression to the B-mode frames using the `log1p()` function as depicted in [Fig. 2](#). This nonlinear scaling method increased the visibility of low-intensity signals while preserving bright regions. Additionally, the Hilbert transform was optionally applied to generate the envelope of RF-like structures, improving contrast and edge visibility. This step simulates real-time ultrasound post-processing and more closely resembles the clinical display quality.

4.2. Data preprocessing

Prior to organizing the dataset for public release, several preprocessing steps are carried out to ensure structural uniformity and enhance visual interpretability while preserving the raw diagnostic nature of the data. Each ultrasound scan, consisting of 100 consecutive grayscale frames, is stored in .mat format with a pixel resolution of 256×256. The data arrays are first processed to remove any singleton dimensions and confirmed to be in the correct 3D shape for downstream tasks.

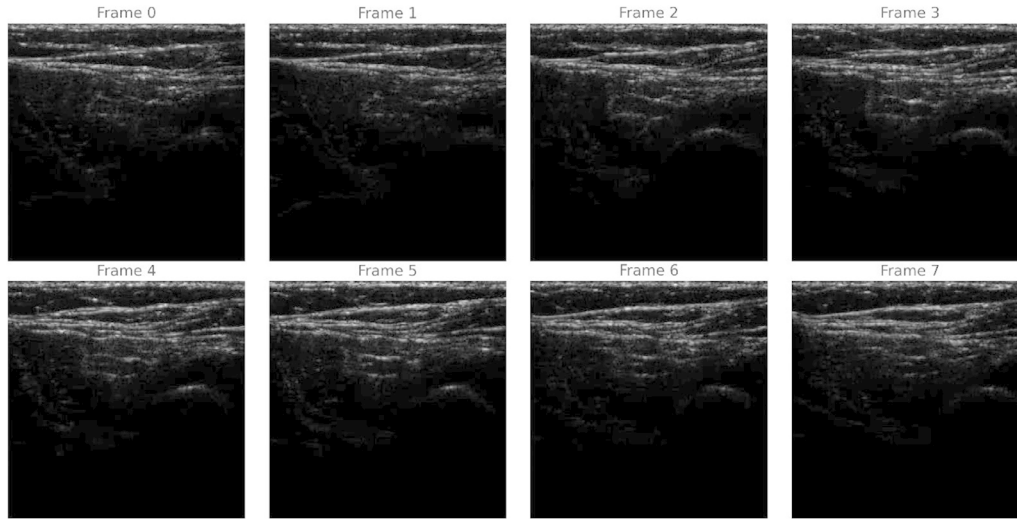


Fig 1. Raw B-mode ultrasound frames acquired before applying log compression and Hilbert transform. These 8 representative frames (Frame 0–7) illustrate the low dynamic range and poor contrast of the original data, which necessitate preprocessing for meaningful visualization and downstream analysis.

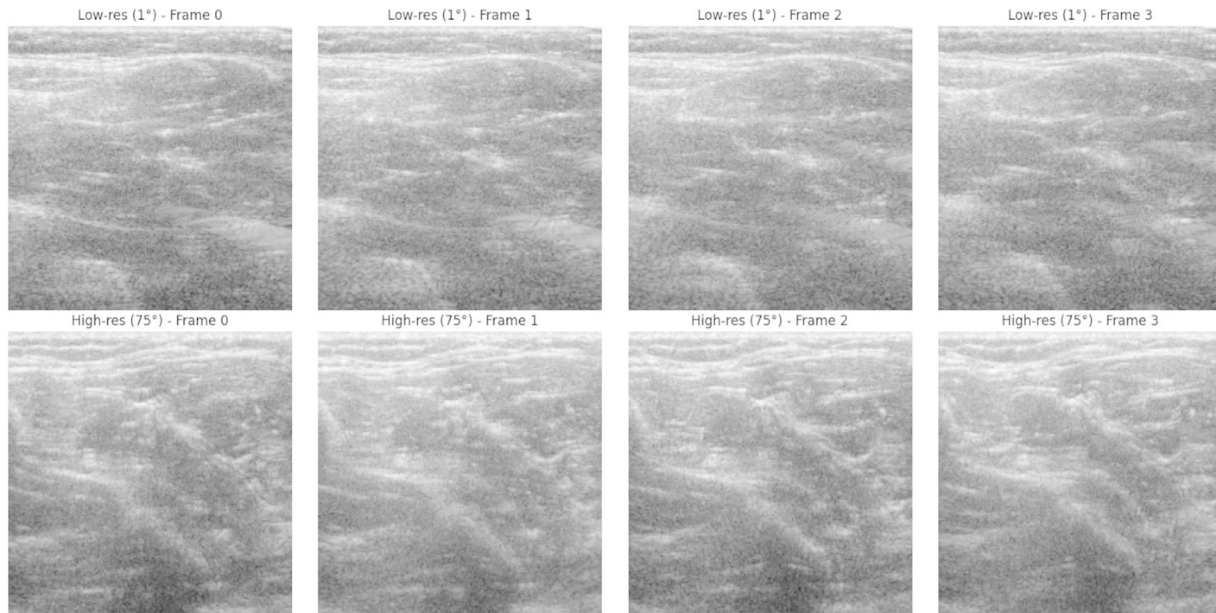


Fig 2. Comparison of B-mode frames after preprocessing using log compression and Hilbert transform. The top row shows low-resolution (1° aperture) images, while the bottom row displays corresponding high-resolution (75° aperture) images. This contrast highlights the enhanced structural visibility and resolution gain from multi-angle compounding.

To address the limited visibility of low-intensity echoes in raw B-mode images, **logarithmic compression** [4] is applied to the amplitude values using the `log1p()` function. This nonlinear transformation improves the dynamic range and makes subtle tissue structures more distinguishable. In selected samples, we have also applied the **Hilbert transform** [5] to simulate envelope detection, enhancing the contrast of tissue interfaces and mimicking real-time ultrasound system outputs.

No additional filters, noise removal techniques, or manual enhancements are applied. The file naming convention is standardized to encode subject ID, arm (left/right), and orientation (longitudinal/transverse), ensuring compatibility with batch processing pipelines and deep learning dataset loaders. The preprocessing preserves the integrity of raw clinical imaging signals while making the dataset suitable for both visual assessment and algorithm training.

4.3. Possible deep learning algorithms

The dataset is structured to support a wide range of deep learning applications in ultrasound image analysis, particularly in image reconstruction and segmentation tasks. Suitable algorithms include:

- A. Image Reconstruction (HR restoration from LR inputs):
 - a. TransCycleGAN: A CycleGAN-based model adapted for unpaired image translation; suitable for generating HR outputs from LR B-mode inputs where paired training is not feasible [1].
 - b. Pix2Pix: A conditional GAN that performs paired image-to-image translation, ideal for learning direct mappings between LR and HR images when paired samples are available [6].
 - c. ResUNet: Combines residual learning with U-Net architecture; effective for reconstructing fine anatomical details while preserving spatial structure [7].
- B. Segmentation (when labels become available):
 - a. U-Net: A widely adopted convolutional architecture in medical image segmentation; supports precise boundary detection using skip connections.
 - b. DeepLabv3+: A deep encoder-decoder model with atrous spatial pyramid pooling; suitable for multiscale context aggregation in ultrasound segmentation [8].
 - c. TransUNet: A hybrid model that combines transformers with U-Net; useful for capturing long-range dependencies and fine-scale tissue variations [9].
- C. Denoising and Image Enhancement:
 - a. Residual CNNs and Autoencoders: Effective for suppressing speckle noise and enhancing contrast without introducing structural artifacts.
 - b. Attention-based Enhancement Networks: Utilize self-attention mechanisms to selectively amplify diagnostically relevant regions while maintaining global context.
- D. Self-supervised or Contrastive Learning:
 - a. PatchNCE or SimCLR-style frameworks: Can exploit HR-LR image pairs for contrastive representation learning without the need for explicit labels [10].

Performance metrics such as **SSIM**, **PSNR**, **MSE**, and **perceptual loss** are recommended for evaluating reconstruction quality, while **Dice coefficient** or **IoU** would be appropriate for segmentation once annotated data is available.

4.4. Future dataset enhancements and research applications

- a. Annotation and Labeling: Future versions will include manually annotated segmentation masks for key tissue structures such as skin, muscle boundaries, and connective tissues. These annotations will enable supervised training of segmentation models and facilitate evaluation using Dice score or IoU metrics.

- b. Pathological Case Inclusion: Expanding the dataset to include scans from patients with musculoskeletal abnormalities, edema, or vascular conditions will improve clinical relevance and support disease classification research.
- c. Multimodal Ultrasound Data: Additional imaging modes such as Doppler (for blood flow analysis) and elastography (for tissue stiffness characterization) will be incorporated, providing a more comprehensive resource for multimodal ultrasound AI research.
- d. Model Benchmarks and Evaluation: Baseline performance metrics of commonly used deep learning models will be published using this dataset, allowing researchers to compare results and validate novel architectures under standardized conditions.
- e. Public Leaderboard and Challenges: A leaderboard for reconstruction or segmentation performance may be hosted in collaboration with AI challenges to stimulate community involvement and algorithm innovation.

These extensions will significantly enhance the dataset's value for research in biomedical image analysis, ultrasound AI, and clinical decision support systems.

Limitations

This dataset is limited to ultrasound B-mode images of healthy adult forearms and does not include any pathological cases or other anatomical regions, which may restrict its use in broader diagnostic model training. Additionally, while the dataset is extensive in frame count and orientation diversity, it does not include manual annotations or segmentation masks, which may be required for supervised learning tasks involving region labeling or structure tracking. Finally, imaging is performed under controlled laboratory settings, which may differ from clinical environments in terms of noise and variability.

Ethics Statement

This dataset has been acquired as part of a study approved by the IRB of Pusan National University (Protocol No. PNU-2025-001) and conducted under ethical approval from the Pusan National University IRB (2023-74-HR), ensuring compliance with human research guidelines. All volunteers provided written informed consent, and the data have been fully anonymized before processing and sharing.

Credit Author Statement

Imrus Salehin: Conceptualization, Data curation, Software, Writing - review & editing. **Nazmul Huda Badhon:** Methodology, Writing - original draft, Software, Validation. **Md Tomal Ahmed Sajib:** Visualization, Investigation. **Nazmun Nessa Moon:** Supervision.

Data Availability

Ultrasound high-resolution (HR) and low-resolution (LR) Hand arm Tissue image data (B-mode) (Original data) (Mendeley Data).

Acknowledgements

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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